

An Introduction to the The Economics of Production Networks

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University of Queensland Lecture

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 - To understand what is happening in an economy, we need to understand what is happening to those **individual production units, how they interact, and how these activities aggregate up**.

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- ▶ The productivity of some individual production unit can be affected by technical progress, government regulations, weather events, labor strikes, collateral financial constraints, etc.

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'In a complex modern economy, there will be a large number of such shifts in any given period, each small in importance relative to total output. There will be much 'averaging out' of such effects across markets. Cancellation of this sort is, I think, the most important reason why one cannot seek an explanation of the general movements we call business cycles in the mere presence, per se, of unpredictability of conditions in individual markets.'

Robert Lucas (1977), Understanding Business Cycles

Why study production networks?

- ▶ Consider an economy with N production units uniformly distributed on $[0, 1]$ with

$$\Delta Z_{jt} = \lambda_j F_t + \epsilon_{jt}^z,$$

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- ▶ If these production units have equal 'size,' $1/N$, aggregate productivity growth is given by

$$\Delta Z_t = N^{-1} \sum_j \Delta Z_{jt} = \bar{\lambda} F_t + N^{-1} \sum_j \epsilon_{jt}^z,$$

where $\bar{\lambda}$ is the average factor loading across units, $N^{-1} \sum_j \lambda_j$.

Why study production networks?

- ▶ $\text{var}(\epsilon_{jt}^z) = \sigma_z^2 \ \forall j \text{ and } t \Rightarrow$ Lucas (1977), the ϵ_{jt}^z 's 'average out' so we only care about aggregate shocks,

$$\lim_{N \rightarrow \infty} \underbrace{\text{var} \left(N^{-1} \sum_j \epsilon_{jt}^z \right)}_{\left(\frac{1}{N^2} \right) N \sigma_z^2} = 0 \quad \Rightarrow \quad \text{var}(\Delta Z_t) = \bar{\lambda}^2 \text{var}(F_t)$$

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- ▶ So then, the [Real Business Cycles](#) research agenda driven by aggregate productivity shocks re-shaped macroeconomics in the 1980s, and over the next 30 years (allowing for other aggregate shocks, monetary policy, government spending, etc.)

The One-Sector Growth Model

$$\max \sum_{t=0}^{\infty} \beta^t \log(C_t)$$

subject to

$$C_t + I_t = Z_t K_t^\alpha L_t^{1-\alpha}, \quad \alpha \in (0, 1),$$

$$K_{t+1} = I_t + (1 - \delta)K_t, \quad \delta \in (0, 1),$$

$$K_0 \geq 0 \text{ given.}$$

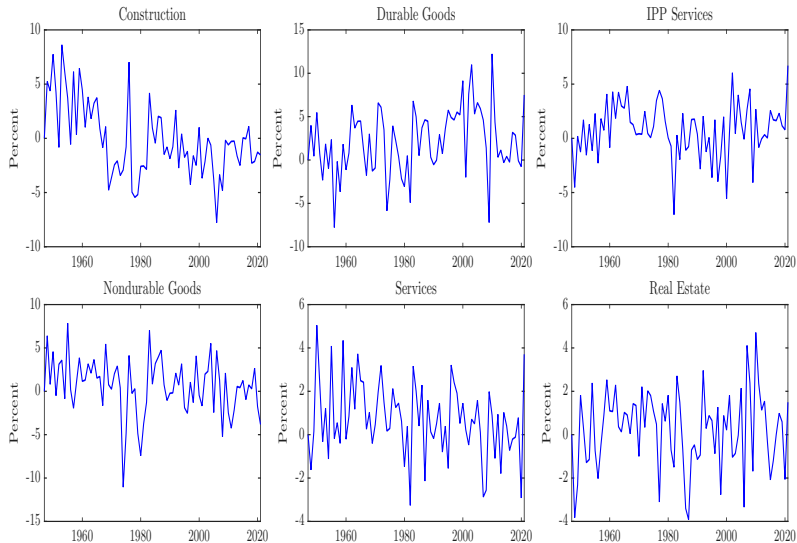
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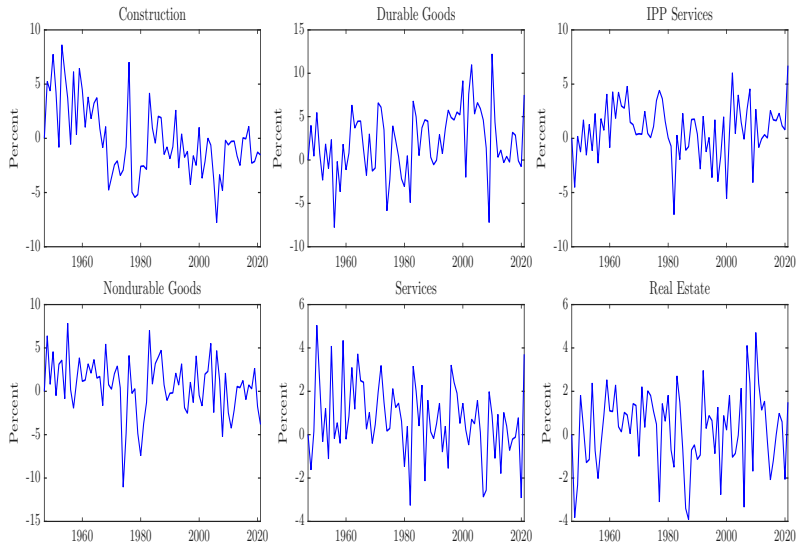
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- ▶ Some exceptions - Long and Plosser (JPE 1983), Dupor (JME 1999), Horvath (JME 2000) - Later, Gabaix (Ecta 2011), Foerster, Sarte, Watson (JPE 2011), Acemoglu, Carvalho, Ozdaglar, Tahbaz-Salehi (Ecta 2012).

Sectoral Productivity Growth



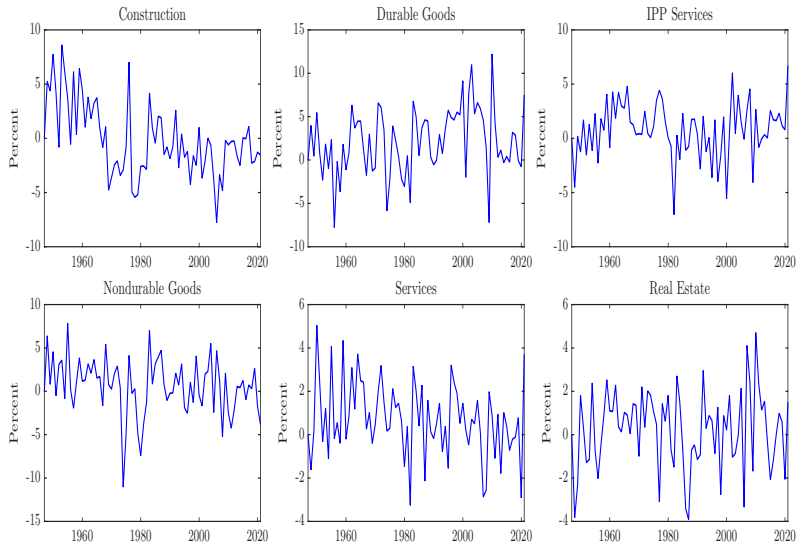
- ▶ A problem with the intuition from Lucas (1977):
- ▶ Disparate experiences and little comovement across sectors, $\lambda_j \approx 0$ for most j , e.g., Foerster et al. (2022).

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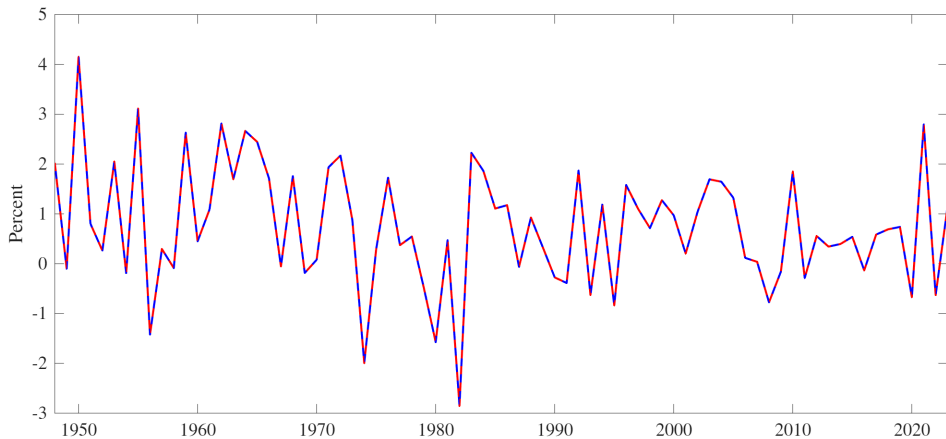
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- ▶ Then $\Delta Z_t \approx N^{-1} \sum_j \epsilon_{jt}^z$, but this implies $\lim_{N \rightarrow \infty} \text{var}(\Delta Z_t) = 0$? ...
- ▶ ... so what explains large aggregate variations in aggregate productivity (or output)?

Aggregate Total Factor Productivity



TFP Summary Statistics by Sector

Sector	<u>Shares</u>		<u>Sectoral Productivity Growth</u>				
	Value-Added	Domar	Mean	Std. Dev.	Min	Max	1st Autocorr
Construction	5.4	12.0	-0.1	1.5	-4.0	3.5	0.5
Durable Goods	14.3	34.8	0.8	1.5	-3.4	5.0	0.1
IPP Services	10.4	16.8	0.5	1.6	-4.7	4.1	0.0
Nondurable Goods	14.2	39.7	0.2	1.2	-3.9	2.9	0.1
Services	42.9	71.8	0.4	1.0	-1.7	2.7	-0.1
Real Estate	12.8	16.9	0.2	1.4	-3.2	3.5	0.1

The Aggregate Effects of Idiosyncratic Disturbances: Hulten (1978)

- ▶ N production units. Gross output is produced using the CRS technology,

$$Y_j = \mathcal{F}^Y(V_j, \mathbf{M}_j),$$

where $\mathbf{M}_j \in \mathcal{R}_+^N$ is a vector of intermediate inputs, $\mathbf{M}_j = (M_{1j}, \dots, M_{Nj})$, where M_{ij} represents the quantity of materials produced in sector i used by sector j (i.e., think $M_{i \rightarrow j}$).

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- ▶ **Exogenous Variables:** Aggregate labor, L , and capital, K , (e.g., land) are supplied inelastically, and Z_j denotes (value added) unit-specific productivity.

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- ▶ Preferences are homothetic $\Rightarrow \exists$ a monotone transformation of U , $\mathcal{U}^{-1}$ that defines a **consumption bundle** $\mathcal{U}^{-1}(U(\mathbf{C})) \equiv C \equiv C(\mathbf{C})$ such that

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- ▶ We can then write

$$U(\mathbf{C}) = \mathcal{U}(C).$$

The Aggregate Effects of Idiosyncratic Disturbances: Hulten (1978)

- ▶ Cobb-Douglas preferences,

$$C = \prod_{j=1}^N \left(\frac{C_j}{\theta_j} \right)^{\theta_j}, \quad \sum_{j=1}^N \theta_j = 1, \quad \theta_j \geq 0 \quad \forall j.$$

with

$$\mathcal{U}(C) = (C^{1-\sigma} - 1)/(1 - \sigma)$$

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$$\frac{\partial C}{\partial C_j} = \frac{\theta_j C}{C_j} \quad \Rightarrow \quad \sum_{j=1}^N C_j \frac{\partial C}{\partial C_j} = C \sum_{j=1}^N \theta_j = C.$$

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- ▶ The resource constraint in each sector, j , is

$$C_j + \sum_i M_{ji} = Y_j \quad \forall j.$$

Solving the Planner's problem

$$\begin{aligned}\max \mathcal{L} = & \mathcal{U}(C) \\ & + \Lambda [C(\mathbf{C}) - C] \\ & + \sum_j \Lambda P_j \left[\mathcal{F}_j^Y(V_j, \mathbf{M}_j) - C_j - \sum_i M_{ji} \right] \\ & + \sum_j \Lambda P_j^V \left[Z_j \mathcal{F}_j^V(K_j, L_j) - V_j \right] \\ & + \Lambda W \left[L - \sum_j L_j \right] \\ & + \Lambda R \left[K - \sum_j K_j \right].\end{aligned}$$

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$$P_j^V Z_j \frac{\partial \mathcal{F}_j^V}{\partial L_j} = W,$$

$$P_j^V Z_j \frac{\partial \mathcal{F}_j^V}{\partial K_j} = R.$$

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- ▶ Here, $C = GDP$.
- ▶ Using the CRS properties of $\mathcal{F}_j^Y$ and the planner's other FOCs, one can show that

$$GDP = \sum_i P_i Y_i - \sum_i \sum_j P_j^Y M_{ji} = \sum_i P_i^V V_i = C.$$

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► Let $x = \log X \Rightarrow Xdx = dX$.

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► The envelope condition gives

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►

$$\frac{\partial(\text{gdp})}{\partial z_j} = \frac{P_j^V V_j}{\text{GDP}} = s_j.$$

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- ▶ Similarly,

$$\frac{\partial \text{gdp}}{\partial k} = \frac{RK}{\text{GDP}} = s^k.$$

Growth Accounting

- ▶ Solving the planner's problem gives $gdp = f(\{Z\}_{j=1}^N; K, L)$. Then

$$d(gdp) = \sum_{j=1}^N \frac{\partial gdp}{\partial z_j} dz_j + \frac{\partial gdp}{\partial k} dk + \frac{\partial gdp}{\partial \ell} d\ell.$$

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- ▶ substituting from the Envelope condition,

$$\begin{aligned} d(gdp) &= \sum_{j=1}^N s_j dz_j + s_d^k k + s^\ell d\ell \\ \Rightarrow dZ &= \sum_{j=1}^N s_j dz_j \quad \text{where } s_j = \frac{P_j^V V_j}{GDP}. \end{aligned}$$

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- ▶ K and L inelastically supplied, so $dk = d\ell = 0$. Then

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- ▶ Gabaix (Ecta 2011):

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- ▶ In practice, $s_j \neq 1/N$, then

$$\lim_{N \rightarrow \infty} \sum_{j=1}^N (s_j)^2 \text{var}(dz_j) = ?$$

Domar Weights vs. Value Added Shares in GDP

- ▶ We had

$$Y_j = \mathcal{F}^Y(V_j, \mathbf{M}_j) \quad \text{and} \quad V_j = Z_j \mathcal{F}^V(K_j, L_j).$$

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- ▶ Repeat the exercise and show that

$$\frac{\partial(\text{gdp})}{\partial z_j} = \frac{P_j Y_j}{\text{GDP}} = d_j.$$

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- ▶ Repeat the exercise and show that

$$\frac{\partial(\text{gdp})}{\partial z_j} = \frac{P_j Y_j}{\text{GDP}} = d_j.$$

- ▶ Note: while $\sum_{j=1}^N s_j = 1$, we have that $\sum_{j=1}^N d_j > 1$.

$$s_j = \frac{P_j^V V_j}{P_j Y_j} d_j.$$

The Aggregate Effects of Idiosyncratic Disturbances

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- ▶ But where does $s_j \forall j$ come from? In what way does the **structure of the economy** determine **unit value added shares**? What if the structure of the economy changes? $\Rightarrow$ Acemoglu, Carvalho, Ozdaglar, Tahbaz-Salehi (Ecta, 2012). Earlier attempt by Dupor (JME 1999)

A Structural Model: Acemoglu, Carvalho, Ozdaglar, Tahbaz-Salehi (2011)

$$\mathcal{U}(C) = \frac{C^{1-\sigma} - 1}{1-\sigma}, \quad \sigma \in [0, \infty)$$

$$C = \prod_{j=1}^N \left(\frac{C_j}{\theta_j} \right)^{\theta_j}, \quad \sum_{j=1}^N \theta_j = 1, \quad \theta_j \geq 0$$

$$V_j = Z_j L_j$$

$$Y_j = \left(\frac{V_j}{\gamma_j} \right)^{\gamma_j} \left(\frac{M_j}{1-\gamma_j} \right)^{(1-\gamma_j)}, \quad \gamma_j \in [0, 1]$$

$$M_j = \prod_{i=1}^N \left(\frac{M_{ij}}{\phi_{ij}} \right)^{\phi_{ij}}, \quad \sum_{i=1}^N \phi_{ij} = 1, \quad \phi_{ij} \geq 0$$

$$C_j + \sum_{i=1}^N M_{ji} = Y_j, \quad \sum_j L_j = \bar{L}$$

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$$\max \mathcal{L} = \frac{C^{1-\sigma} - 1}{1-\sigma} \quad (\text{P1})$$

$$+ \Lambda \left[\prod_{j=1}^N \left(\frac{C_j}{\theta_j} \right)^{\theta_j} - C \right]$$

$$+ \sum_{j=1}^N \Lambda P_j \left[\left(\frac{V_j}{\gamma_j} \right)^{\gamma_j} \left(\frac{M_j}{1-\gamma_j} \right)^{1-\gamma_j} - C_j - \sum_{i=1}^N M_{ji} \right]$$

$$+ \sum_{j=1}^N \Lambda P_j^V [Z_j L_j - V_j]$$

$$+ \sum_{j=1}^N \Lambda P_j^M \left[\prod_{i=1}^N \left(\frac{M_{ij}}{\phi_{ij}} \right)^{\phi_{ij}} - M_j \right]$$

$$+ \sum_{j=1}^N \Lambda W \left[\bar{L} - \sum_j L_j \right].$$

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 \end{aligned}$$

$$C^{-\sigma} = \Lambda$$

$$\frac{\theta_j C}{C_j} = P_j, \quad \forall j$$

$$1 = \prod_{j=1}^N (P_j)^{\theta_j}$$

$$\gamma_j \frac{P_j Y_j}{V_j} = P_j^V$$

$$(1-\gamma_j) \frac{P_j Y_j}{M_j} = P_j^M, \quad \forall j$$

$$\frac{P_j^V V_j}{L_j} = W, \quad \forall j$$

$$\phi_{ij} \frac{P_j^M M_j}{M_{ij}} = P_i, \quad \forall i, j$$

National Income Accounting

►

$$GDP = \sum_{j=1}^N P_j C_j = C.$$

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$$\begin{aligned} C &= \sum_j P_j Y_j - \sum_j \sum_i P_j M_{ji}, \\ &= \sum_j P_j^V V_j \\ &= \sum_j W L_j \\ &= W \bar{L}. \end{aligned}$$

Equilibrium Prices and the Leontief Inverse

- ▶ Let $\mathbf{Z} = (Z_1, \dots, Z_N)'$, and define $\mathbf{Y} = (Y_1, \dots, Y_N)'$, $\mathbf{V} = (V_1, \dots, V_N)'$, $\mathbf{P} = (P_1, \dots, P_N)'$, etc. Let $\mathbf{\Gamma}_d = \text{diag}[\gamma]_j$, $\mathbf{\Phi} = [\phi_{ij}]$, etc.

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- ▶ From the equilibrium conditions,

$$\left(\frac{\gamma_j}{V_j}\right)_j^\gamma (P_j Y_j)^{\gamma_j} \left(\frac{1-\gamma_j}{M_j}\right)^{1-\gamma_j} (P_j Y_j)^{1-\gamma_j} = P_j = (P^V)^{\gamma_j} (P^M)^{1-\gamma_j}$$
$$\Rightarrow \log \mathbf{P} = \mathbf{\Gamma}_d \log \mathbf{P}^V + (\mathbf{I} - \mathbf{\Gamma}_d) \log \mathbf{P}^M.$$

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▶

$$P_i = \frac{\phi_{ij}}{M_{ij}} P_j^M M_j \Rightarrow \log \mathbf{P}^M = \mathbf{\Phi}' \log \mathbf{P}.$$

Equilibrium Prices and the Leontief Inverse



$$\log \mathbf{P}^V = \mathbf{\Gamma}_d^{-1} (\mathbf{I} - (\mathbf{I} - \mathbf{\Gamma}_d)\mathbf{\Phi}') \log \mathbf{P}$$

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►

$$\log \mathbf{P} = \underbrace{(\mathbf{I} - (\mathbf{I} - \Gamma_d)\Phi')^{-1}}_{\mathcal{L}, \text{ Leontief Inverse}} \Gamma_d (\mathbf{1} \log W - \log \mathbf{Z})$$

What is a Leontief Inverse?

- ▶ Let

$$\mathbf{A} = (\mathbf{I} - \mathbf{\Gamma}_d)\mathbf{\Phi}' \text{ so that } \mathcal{L} = (\mathbf{I} - \mathbf{A})^{-1}.$$

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- ▶ Let

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- ▶ $\mathbf{A}$ is element-wise non-negative and its column sums are strictly less than one. In the special case, $\mathbf{\Gamma}_d = \gamma \mathbf{I}$,

$$\begin{aligned}\mathbf{A} &= (1 - \gamma)\mathbf{\Phi}' \\ \Rightarrow \sum_{j=1}^N a_{ji} &= (1 - \gamma) \sum_{j=1}^N \phi_{ij} < 1.\end{aligned}$$

The columns of $\mathbf{A}$ correspond to the (weighted) rows of $\mathbf{\Phi}$.

What is a Leontief Inverse?

- ▶ The spectral radius of $\mathbf{A}$, $\rho(\mathbf{A})$, is strictly less than one,

$$\rho(\mathbf{A}) = \max_i \{\lambda_i(\mathbf{A})\} < 1, \quad \lambda_i(\mathbf{A}) \text{ is the } i^{th} \text{ eigenvalue of } \mathbf{A}.$$

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- ▶ $\rho(\mathbf{A}) < 1 \Rightarrow \mathbf{I} - \mathbf{A}$ is an invertible M-matrix $\Rightarrow$ it has an element-wise non-negative inverse (Plemmons, 1977, Theorem 2),

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- ▶ In addition, $\rho(\mathbf{A}) < 1 \Rightarrow$

$$\mathcal{L} = (\mathbf{I} - \mathbf{A})^{-1} = \sum_{k=0}^{\infty} \mathbf{A}^k \quad \text{- Neumann Series}$$

- ▶ ℓ_{ij} , the $(i, j)^{\text{th}}$ element of $\mathcal{L}$, measures the importance of industry j as a direct and indirect supplier to industry i . Industry j produces output used by industry i , which produces output used by industry j , and so on for all industry pairs ...

Hulten's Theorem Redux

- ▶ $\log \mathbf{P} = \mathcal{L}\Gamma_d(\mathbf{1} \log W - \log \mathbf{Z})$

$$\Rightarrow \log \mathbf{P} = \mathbf{1} \log W - \mathcal{L}\Gamma_d \log \mathbf{Z} \quad \text{because } \mathcal{L}\Gamma_d \mathbf{1} = \mathbf{1}.$$

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- ▶ Recall that $\theta' \log \mathbf{P} = 0$ and $\theta' \mathbf{1} = 1$. Then

$$\begin{aligned} \log W &= \theta' \mathcal{L}\Gamma_d \log \mathbf{Z}, \\ \Rightarrow \log W + \log \bar{L} &= \theta' \mathcal{L}\Gamma_d \log \mathbf{Z} + \log \bar{L}, \quad \text{- recall that } \bar{L}, \text{ is exogenous,} \end{aligned}$$

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$$\log(GDP) = \log \bar{L} + \theta' \mathcal{L}\Gamma_d \log \mathbf{Z} \Rightarrow \frac{\partial \log(GDP)}{\partial \log Z_j} = [\theta' \mathcal{L}\Gamma_d]_j.$$

Aggregation Irrelevance: Dupor (1999)

- ▶ From the last slide,

$$\log(GDP) = \log \bar{L} + \underbrace{\log \mathbf{Z}' \Gamma_d}_{\epsilon^{z'}} \mathcal{L}' \theta \quad \text{where } \mathcal{L} = (\mathbf{I} - (\mathbf{I} - \Gamma_d) \Phi)^{-1}.$$

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- Suppose that $\theta_j = 1/N \forall j$, that $\gamma_j = \gamma \forall j$, and $\Phi \mathbf{1} = \lambda \mathbf{1}$. Then

$$(\mathbf{I} - (\mathbf{I} - \Gamma_d) \Phi)^{-1} \mathbf{1} = \frac{1}{1 - (1 - \gamma) \lambda} \mathbf{1},$$

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- and

$$\lim_{N \rightarrow \infty} \text{var}(\log(GDP)) = 0.$$

Hulten's Theorem Redux

- ▶ We established that $\log(GDP) = \log \bar{L} + \boldsymbol{\theta}' \mathcal{L} \boldsymbol{\Gamma}_d \log \mathbf{Z}$ and $\frac{\partial \log(GDP)}{\partial \log Z_j} = [\boldsymbol{\theta}' \mathcal{L} \boldsymbol{\Gamma}_d]_j$. But what is $[\boldsymbol{\theta}' \mathcal{L} \boldsymbol{\Gamma}_d]_j$?

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▶

$$P_j^Y C_j = P_j^Y Y_j - \sum_i P_j^Y M_{ji}$$

$$\Rightarrow \theta_j C = \frac{1}{\gamma_j} P_j^V V_j - \sum_i \phi_{ji} \frac{(1 - \gamma_i)}{\gamma_i} P_i^V V_i$$

$$\Rightarrow \theta C = (I - \Phi(I - \Gamma_d)) \Gamma_d^{-1} (P^V \circ V)$$

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▶

$$\frac{P^V \circ V}{C} = \Gamma_d (I - \Phi(I - \Gamma_d))^{-1} = \Gamma_d \mathcal{L}' \theta = s.$$

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$$\frac{\partial \log(GDP)}{\partial \log Z_j} = [\boldsymbol{\theta}' \boldsymbol{\mathcal{L}} \boldsymbol{\Gamma}_d]_j = s_j.$$

- ▶ This is Hulten's theorem, but we can now relate the **vector of value-added shares in GDP** (or relative size of production units), $\mathbf{s}$, to the structure of the economy:
 - $\boldsymbol{\Gamma}_d$, sectoral value added shares in gross output,
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Hulten's Theorem Redux

- ▶ So we have

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- ▶ These features of the economy interact to jointly determine the relative size distribution of sectors, i.e., the distribution of the elements of $\mathbf{s}$.

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The Aggregate Implications of Idiosyncratic Disturbances

- Suppose $\theta_j = 1/N \forall j$. Then

$$\begin{aligned}\log(GDP) &= \log \bar{L} + \frac{1}{N} \sum_{i=1}^N \underbrace{\left(\sum_{j=1}^N \ell_{ji} \right)}_{c_i} \gamma_i \log Z_i \\ &= \log \bar{L} + \frac{1}{N} \sum_{i=1}^N c_i \underbrace{\gamma_i \log Z_i}_{\epsilon_i^Z}\end{aligned}$$

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- Since the Leontief inverse satisfies $\mathcal{L} = \mathbf{I} + \mathcal{L}\mathbf{A}$, where $\mathbf{A} = (\mathbf{I} - \mathbf{\Gamma}_d)\mathbf{\Phi}'$

$$c_i = 1 + \sum_{j=1}^N c_j a_{ji}$$

c_i is the **Bonacich centrality** with $(\alpha = 0, \beta = 1)$ of vertex i in the economy's production network.

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- ▶ c_i is larger, i.e., industry i is more **central**, if i is a key supplier to other central industries.

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- ▶ We established

$$\log(GDP) = \log \bar{L} + \frac{1}{N} \sum_{i=1}^N c_i \epsilon_i^z \quad \text{where } \epsilon_i^z = \gamma_i \log Z_i.$$

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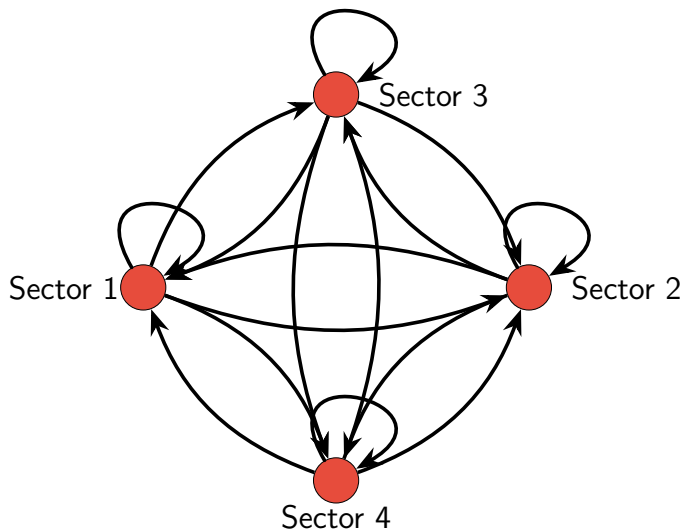
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A Dense Production Network



Aggregation Irrelevance in a Dense Network

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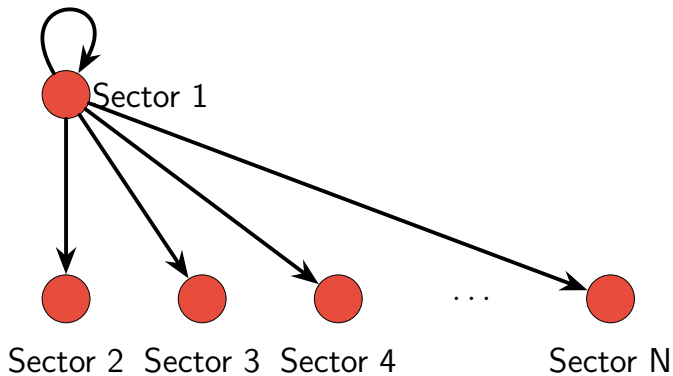
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One Dominant Industry: The Star Network



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$$\lim_{N \rightarrow \infty} \sigma_{GDP}^2 = \frac{(1 - \gamma)^2}{\gamma^2} > 0$$

Further Topics in Production Networks

- ▶ Escaping Hulten's Theorem
 - Non-linearities in production, e.g., Baqaee and Farhi (2019)
 - Elastic factor supply, e.g., von Lehn and Winberry (2022)
- ▶ Distortions in production networks, e.g., Baqaee and Farhi (2020)
- ▶ Production networks in dynamic economies, e.g. Long and Plosser (JPE 1983), and many others ...
- ▶ Long-run implications of production network, balanced growth and long run trends, e.g., Foerster, Sarte, Watson (2022), structural change, etc.
- ▶ Production networks across countries, implications for economic development, e.g., Gloria, Miranda-Pinto, Muñoz-Flemming (2024)
- ▶ ...